Glitchwave 567 — MAIN board

Assembly instructions for PCBWay

Customer: Illicit Apothecary
Project: Glitchwave 567 guitar pedal — MAIN board
Board file base name: `glitchwave567`
Revision date of this package: 2026-08-02
Quantity: 5 assembled boards, full turnkey

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1. Files in this submission

File	What it is
`glitchwave567_main_gerbers.zip`	Gerbers (RS-274X) + Excellon drill, PTH and NPTH separate, plus drill maps
`BOM_main_pcbway.csv`	Bill of materials, 96 lines, every line carries an LCSC part number
`CENTROID_main_pcbway.csv`	Pick-and-place, 296 placements, all top side
this PDF	Assembly notes

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2. Board specification

Parameter	Value
Size	138 x 114 mm, with 11 x 11 mm notches at the corners (in Edge.Cuts)
Layers	4
Stackup, top to bottom	L1 `F_Cu` signal / L2 `In1_Cu` **solid ground plane** / L3 `In2_Cu` signal / L4 `B_Cu` signal
Thickness	1.6 mm
Copper	1 oz all layers
Minimum track / clearance used	0.25 mm / 0.13 mm — please build to 5/5 mil, not 6/6
Vias	0.5 mm pad, 0.3 mm drill
Surface finish	HASL lead free
Solder mask / silkscreen	green / white

Please confirm the stackup order above against the gerber file names. L2 must be the solid ground plane.

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3. Things that will otherwise generate a query

3.1 Via in pad

Approximately **246 vias sit inside SMD pads** (0.3 mm drill, 0.5 mm pad). These are ground and decoupling stitching vias down to the L2 plane. Please quote this two ways:

1. as drawn, vias tented, customer accepting the wicking risk; and
2. vias resin filled and capped, plated over (VIPPO).

Customer will choose on receipt of the quote. Do not proceed on assumption.

3.2 Do-not-populate

C41 and C42 must be left unpopulated. The pads are deliberately empty — they are a tuning option in the filter around the LM567 and populating them changes the sound of the product. They are marked DNP in the BOM. Please do not helpfully fit them.

3.3 Raspberry Pi Pico, U20

U20 is a genuine Raspberry Pi Pico module, surface mounted flat onto the board using its castellated edge pads and the official land pattern. It is not a bare RP2040. No firmware loading is required — the customer flashes it over USB after delivery.

3.4 Exposed pad

U19 is an MP1584EN in SOIC-8 with an exposed thermal pad. The pad is electrically connected and must be soldered down.

3.5 No substitutions

Every BOM line carries a specific LCSC part number, verified in stock. **Please do not substitute any part**, including passives, without written approval. This is an analog audio product and part-to-part differences are audible.

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4. Assembly

- All components are **top side only**. Nothing is fitted to the bottom.
- 272 SMD placements, 23 through-hole placements, 84 distinct part numbers.
- No BGA, QFN or QFP parts. No X-ray inspection required.
- J10 is a 2x8 female socket on the **top** side. It mates with a male header on the CONTROL board and is the mechanical and electrical link between the two boards. Its height matters: 8.50 mm body. Do not substitute a different socket height.
- Boards ship as single pieces, not panelised.

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5. Contact

Questions on this package should go back to the customer before production starts, not after.